2026

UK Crime Data
Exploratory Data Analysis

EDA REPORT

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1. Executive Summary

The analysis identifies clear differences in crime patterns across the selected police forces, driven primarily by variations in population density, urban structure, and geographic coverage. Metropolitan areas, particularly the Metropolitan Police Service, exhibit significantly higher crime volumes and strong spatial concentration of incidents within central urban zones. In contrast, regions such as Surrey and Thames Valley show more dispersed crime patterns, reflecting their suburban and semi-rural characteristics.

When adjusting for population size, the findings indicate that high crime volume does not necessarily correspond to the highest crime rate. Police forces such as West Midlands and West Yorkshire demonstrate comparatively elevated crime rates, suggesting a greater relative crime burden despite lower total incident counts than London.

Across all regions, crime composition remains broadly consistent, with violence and sexual offences forming the largest proportion of recorded incidents. Temporal trends show relatively stable crime levels over the observed period, with only moderate seasonal variation rather than sustained increases or decreases.

Based on these findings, the Metropolitan Police Service and West Midlands Police are recommended for further investigation. These areas present the most distinct and analytically valuable patterns, enabling meaningful comparison between different urban crime environments.

2. Introduction

This report presents an exploratory data analysis (EDA) of crime data across six UK police forces: City of London Police, Metropolitan Police Service, West Midlands Police, West Yorkshire Police, Thames Valley Police, and Surrey Police. These datasets were selected to provide a diverse and representative comparison of crime patterns across different types of regions within England.

The selected police forces capture a range of geographic and demographic contexts. For example, the Metropolitan Police Service and City of London Police represent highly urbanised areas with high population density and economic activity, while West Midlands and West Yorkshire Police represent large metropolitan regions with multiple urban centres. In contrast, Thames Valley and Surrey Police cover more suburban and semi-rural areas, allowing comparison with less densely populated regions.

By including police forces with varying population sizes, urban structures, and geographic coverage, the analysis enables a more meaningful comparison of crime patterns across different environments. This is particularly important when examining both total crime counts and crime rates per 1000 residents, as these measures can vary significantly depending on regional characteristics.

The dataset consists of police-recorded crime incidents over two years, combined with population data to support standardised comparisons. Before analysis, the data was cleaned and preprocessed to ensure consistency across all datasets.

This analysis aims to identify key patterns in crime levels, crime types, temporal trends, and spatial distribution across these regions. The findings will be used to recommend specific police forces for further investigation in the next stage of the project.

3. Data Dictionary

3.1. Clean Crime Dataset

Column Name	Data Type	Description
crime_id	string	Unique identifier for each crime (may be missing for some records)
month	datetime	Month when the crime was recorded (YYYY-MM format)
reported_by	category	Police force that recorded the crime
falls_within	category	Police force jurisdiction
longitude	float	Geographic longitude of the crime location
latitude	float	Geographic latitude of the crime location
location	string	Description of the crime location (e.g. "On or near High Street")
lsoa_code	category	Lower Layer Super Output Area code (geographic identifier)
lsoa_name	category	Name of the LSOA area
crime_type	category	Type of crime (e.g. burglary, theft, violence)
last_ouctome_category	category	Outcome of the crime (e.g. investigation complete, no suspect identified)
year	integer	Extracted from month
month_num	integer	Numeric month (1–12)

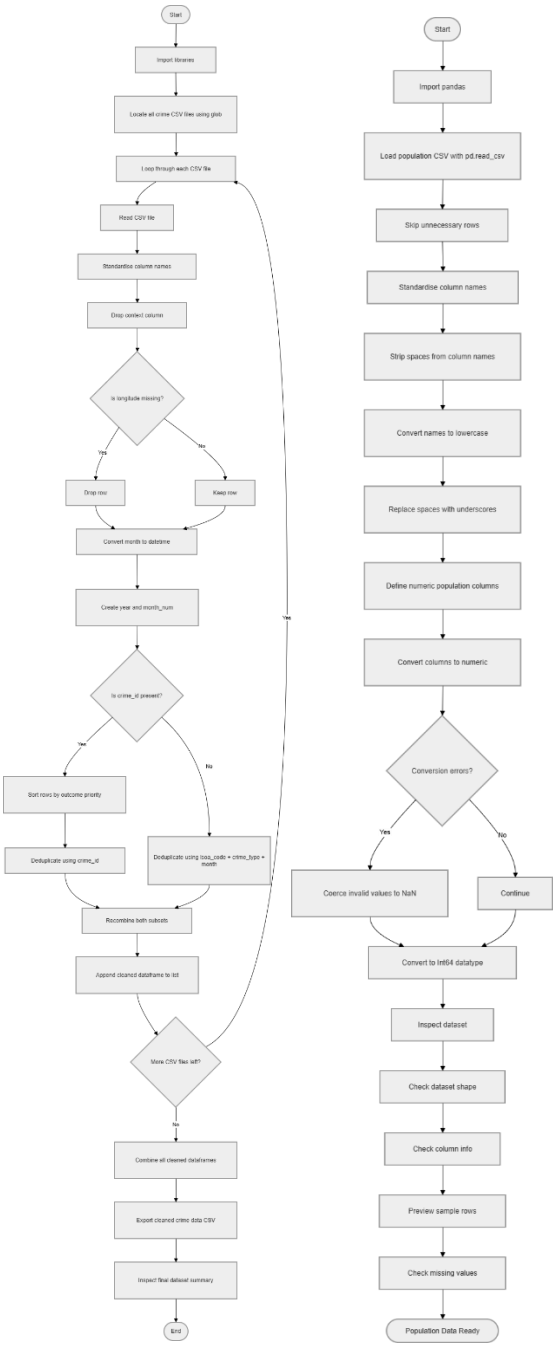
3.2. Clean Population Dataset

Column Name	Data Type	Description
lsoa_code	string	Unique identifier for each Lower Layer Super Output Area (LSOA)
lsoa_name	string	Name of the LSOA area
total	Int64	Total population in the area
f0_to_15	Int64	Female population aged 0–15
f16_to_29	Int64	Female population aged 16–29
f30_to_44	Int64	Female population aged 30–44
f45_to_64	Int64	Female population aged 45–64
f65_and_over	Int64	Female population aged 65+
m0_to_15	Int64	Male population aged 0–15
m16_to_29	Int64	Male population aged 16–29
m30_to_44	Int64	Male population aged 30–44
m45_to_64	Int64	Male population aged 45–64
m65_and_over	Int64	Male population aged 65+

4. Data Cleaning/Preprocessing

Before conducting the exploratory data analysis, a structured preprocessing workflow was developed to ensure that the crime and population datasets were accurate, consistent, and suitable for analysis. Due to the size and complexity of the police crime datasets, preprocessing was first explored on a single police force dataset in a separate preprocessing exploratory notebook. This allowed the cleaning logic, data quality checks, and transformation steps to be tested on a smaller subset before being scaled into a repeatable pipeline for all selected police forces.

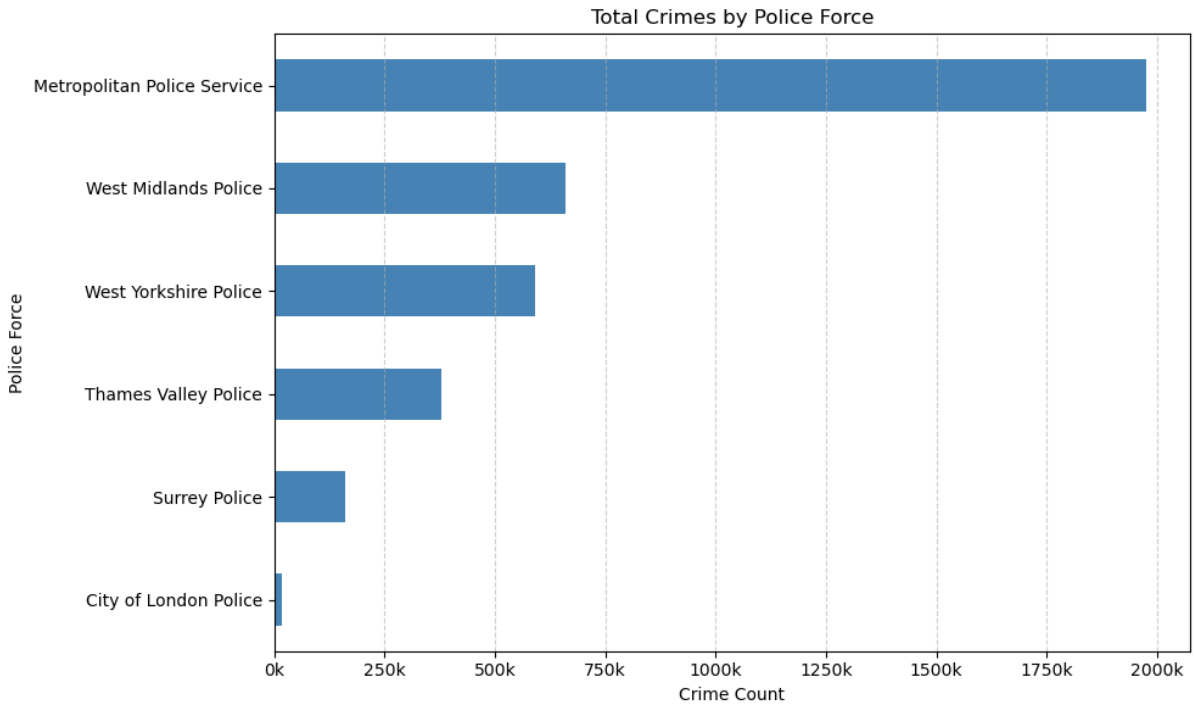
This staged approach improved reliability and reduced the risk of applying incorrect transformations across the full dataset. Once the workflow had been validated, it was implemented in the main preprocessing notebook and applied consistently across all selected crime CSV files.



Preprocessing of Crime and Population Dataset respectively.

5. Exploratory Data Analysis – Number of Crimes

5.1. Number of Crimes in each Police Force Jurisdiction



The Metropolitan Police Service records by far the highest number of crimes, with almost two million incidents, significantly exceeding all other police forces included in the analysis. This is expected as the Metropolitan Police covers Greater London, which has the largest population and one of the most densely populated urban areas in the UK.

West Midland and West Yorkshire Police record the next highest crime totals, with approximately 700,000 and 600,000 incidents respectively. These regions include major metropolitan areas such as Birmingham, Leeds, and Bradford, which tend to experience higher volumes of crime due to larger populations and greater urban activity.

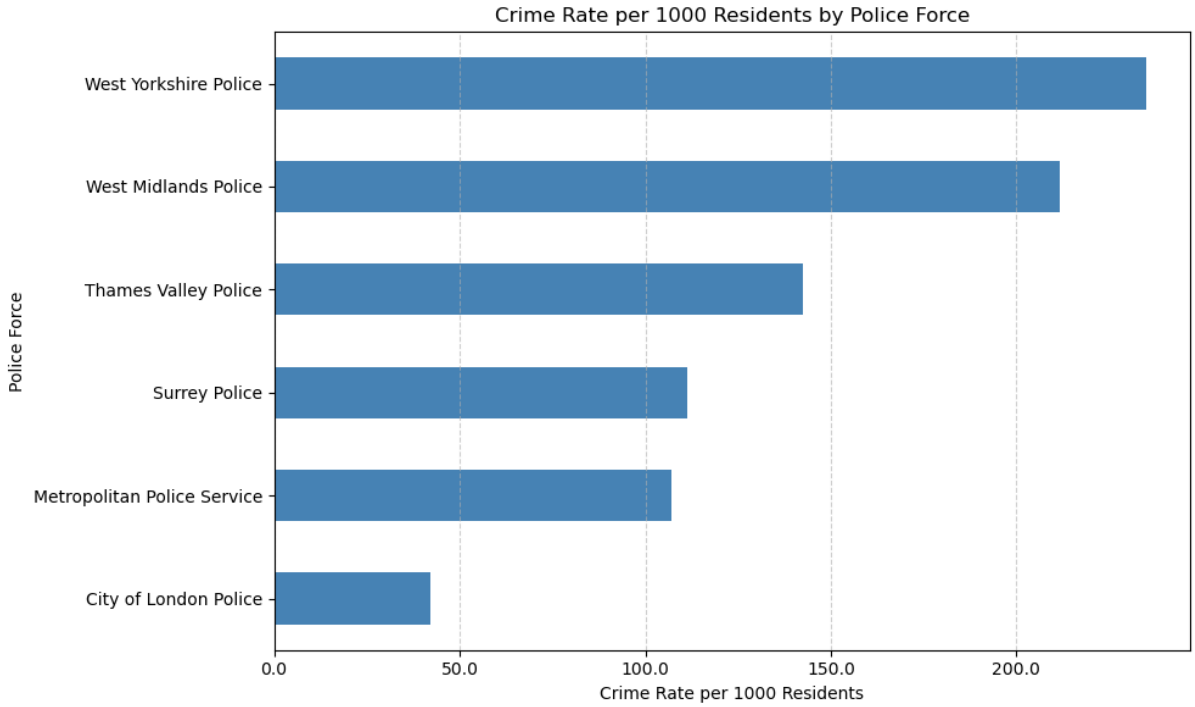
Thames Valley Police record a moderate level of crime incidents, reflecting the mix of urban and suburban environments within its jurisdiction.

Surrey Police records significantly fewer crimes, which may be attributed to its lower population density and more suburban character compared with the larger metropolitan regions.

City of London Police show the lowest total crime count. This is expected because the City of London covers a very small geographic area compared with other police forces.

This chart highlights that total crime counts are heavily influenced by population size and area coverage, meaning they do not necessarily represent the underlying crime risk in each region.

5.2. Crime rate in each Police Force Jurisdiction



This chart standardises crime counts relative to population size. This allows for more meaningful comparison of crime risk across regions.

When adjusting for population, the ranking of police forces changes significantly. West Yorkshire Police have the highest crime rate per 1000 residents, indicating that residents in this region experience a higher relative level of crime compared with other areas.

West Midlands Police also show a relatively high crime rate, reflecting the challenges often associated with large urban environments.

Thames Valley Police and Surrey Police display moderate crime rates, suggesting a more balanced crime environment relative to population.

Interestingly, despite recording the largest number of total crimes, the Metropolitan Police Service has a relatively higher moderate crime rate when adjusted for population. This indicates that the high crime count in London is largely driven by the city's very large population rather than disproportionately high crime risk.

City of London Police again record the lowest crime rate in the dataset. However, this may be influence by the unique demographic characteristics of the City of London, where the daytime population is significantly larger than the residential population.

This comparison highlights the importance of analysing crime rates rather than only absolute crime counts, as it provides a more accurate representation of crime exposure for residents.

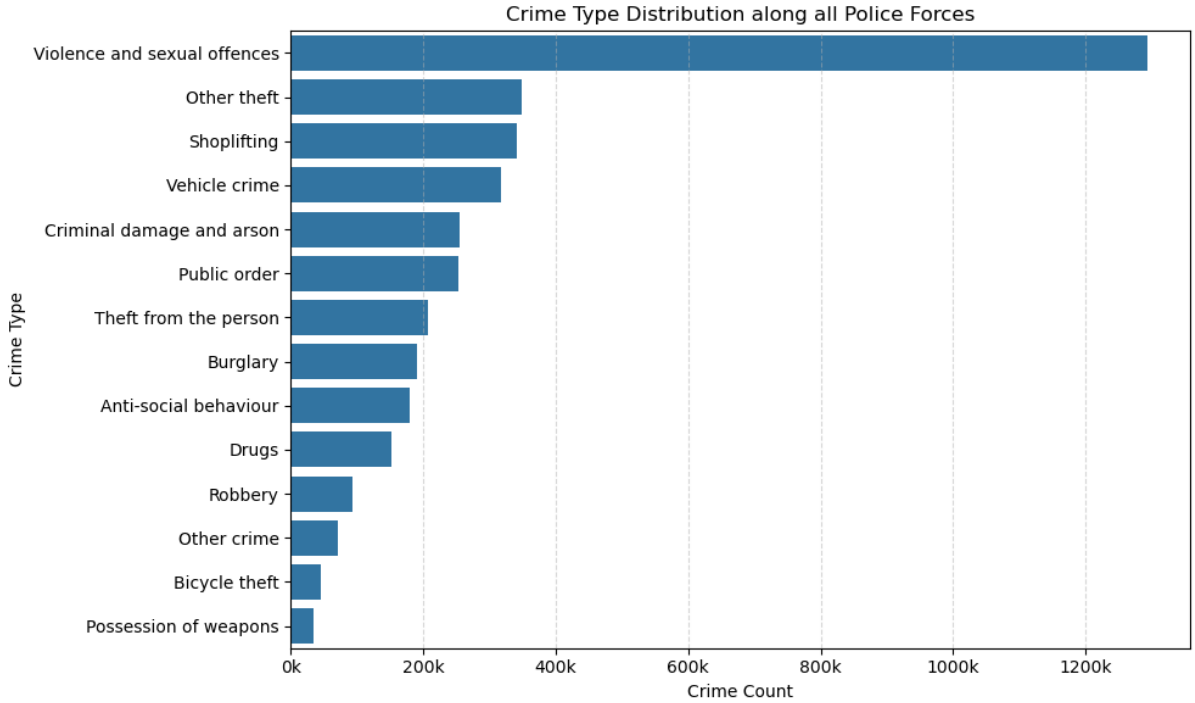
5.3. Number of crimes - Key Insights

Comparing total crime counts with population-adjusted crime rates reveals that regions with the highest number of recorded crimes do not necessarily have the highest crime risk. While the Metropolitan Police report the largest volume of crime incidents, West Yorkshire and West Mdilands exhibit higher crime rates relative to their population. This highlights the importance of normalising crime data when comparing different regions.

Thes findings are relevant for property market analysis, as crime rates can influence perceptions of safety and desirability in residential areas. Regions with higher crime rates may be perceived as less desirable for property investment, while areas with lower relative crime exposure may present stronger opportunities of residential development.

6. Exploratory Data Analysis – Crime Type

6.1. Crime Type Distribution along all Police Forces Jurisdiction



The results indicate that violence and sexual offences are by far the most frequently recorded crimes, significantly exceeding all other categories. This category accounts for a substantially larger number of incidents than any other crime type in the dataset, suggesting that interpersonal offences represent a major component of recorded crime in the analysed regions.

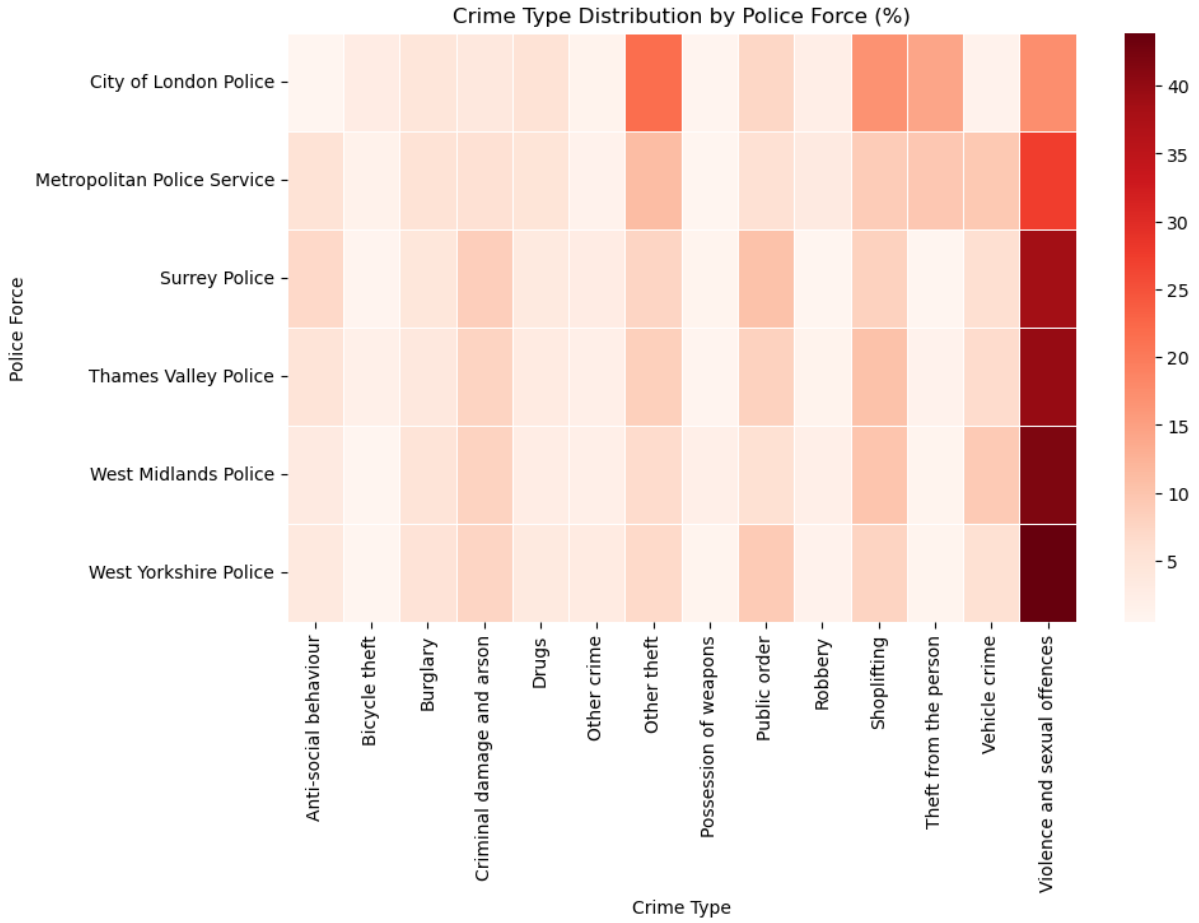
Following this category, other theft and shoplifting represent the next most common crime types. These crimes are typically associated with urban environments and commercial areas, where higher population density and retail activity increase opportunities for theft-related offences.

Vehicle crime and criminal damage and arson also appear frequently in the dataset, indicating that property-related crimes remain an important component of the overall crime landscape.

In contrast, crimes such as bicycle theft, robbery and possession of weapons occur far less frequently. These categories represent a relatively small proportion of the total recorded crimes across all police forces.

Overall, the distribution, suggests that crime incidents are heavily concentrated in a small number of dominant categories, particularly violence-related offences and theft-related crimes.

6.2. Crime Type Distribution by Police Force Jurisdiction



The chart presents a heatmap showing the percentage distribution of crime types within each police force, allowing comparison of crime composition across regions.

Across all police forces, violence and sexual offences consistently represent the largest proportion of recorded crimes, indicating that this crime category is dominant regardless of regional differences. However, the relative contribution of other crime types varies between police forces.

For example, shoplifting and theft-related offences appear more prominent in police forces covering larger metropolitan areas, such as the Metropolitan Police and West Midlands police. This may reflect the presence of large retail districts and higher population density within these regions.

Meanwhile, police forces covering more suburban and semi-urban areas, such as Surrey Police and Thames Valley Police, show a slightly more balanced distribution of crime categories, with theft, public order, and criminal damage contributing relatively similar proportions.

The heatmap also highlights that certain crime types maintain relatively consistent proportions across regions, suggesting common underlying crime patterns throughout the analysed police forces.

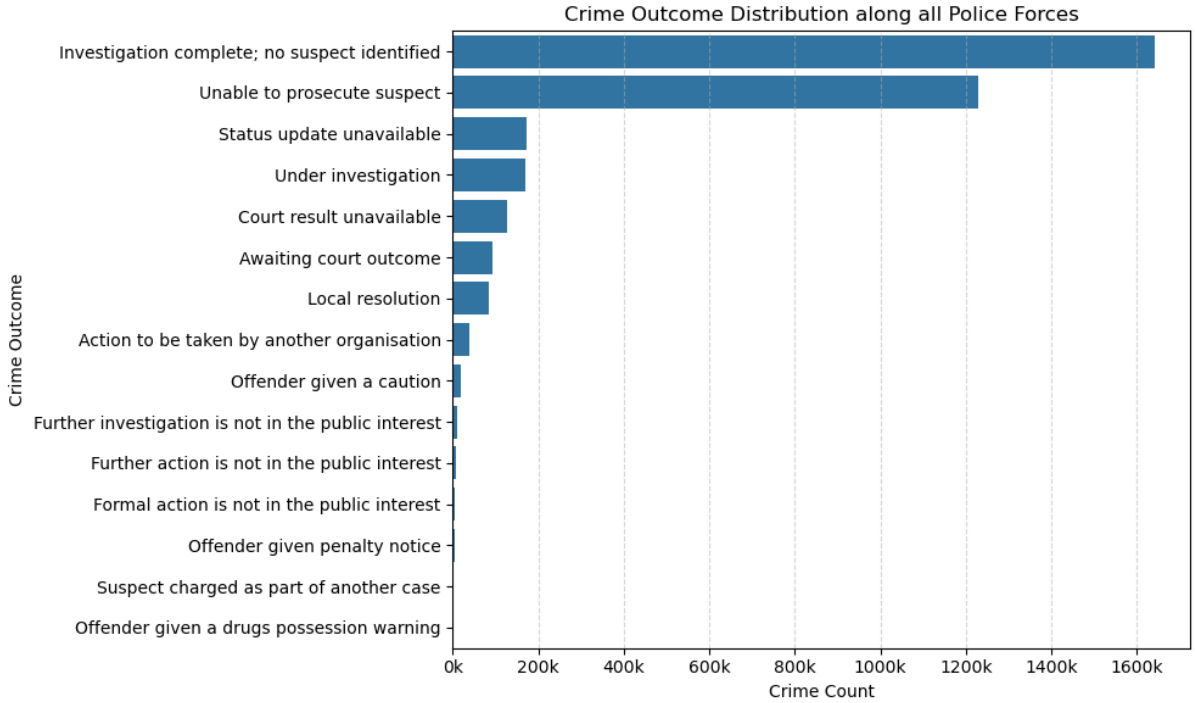
6.3. Crime Type - Key Insights

Taken together, these figures indicate that while total crime volumes vary significantly between police forces, the composition of crime types remains broadly similar across regions. Violence-related offences and theft-related crimes dominate across all police forces, while less common offences such as weapons possession and bicycle theft represent a small share of total incidents.

Understanding the distribution of crime types provides important context for interpreting regional crime patterns and identifying areas where types of offences may require greater attention. Also, it is known that violent crimes usually have a greater negative impact than other crime types on real estate. A deeper analysis is to be made in a later stage, although this shows the importance of analysing crime types in a region.

7. Exploratory Data Analysis – Crime Outcome

7.1. Crime Outcome along all Police Force Jurisdictions



The data shows the distribution of crime outcomes aggregated across all analysed police forces. The results indicate that the most common outcome is "investigation complete: no suspect identified", which accounts for the largest proportion of recorded outcomes. This suggests that a significant number of cases are closed without identifying a suspect, which may reflect challenges in obtaining sufficient evidence or identifying offenders.

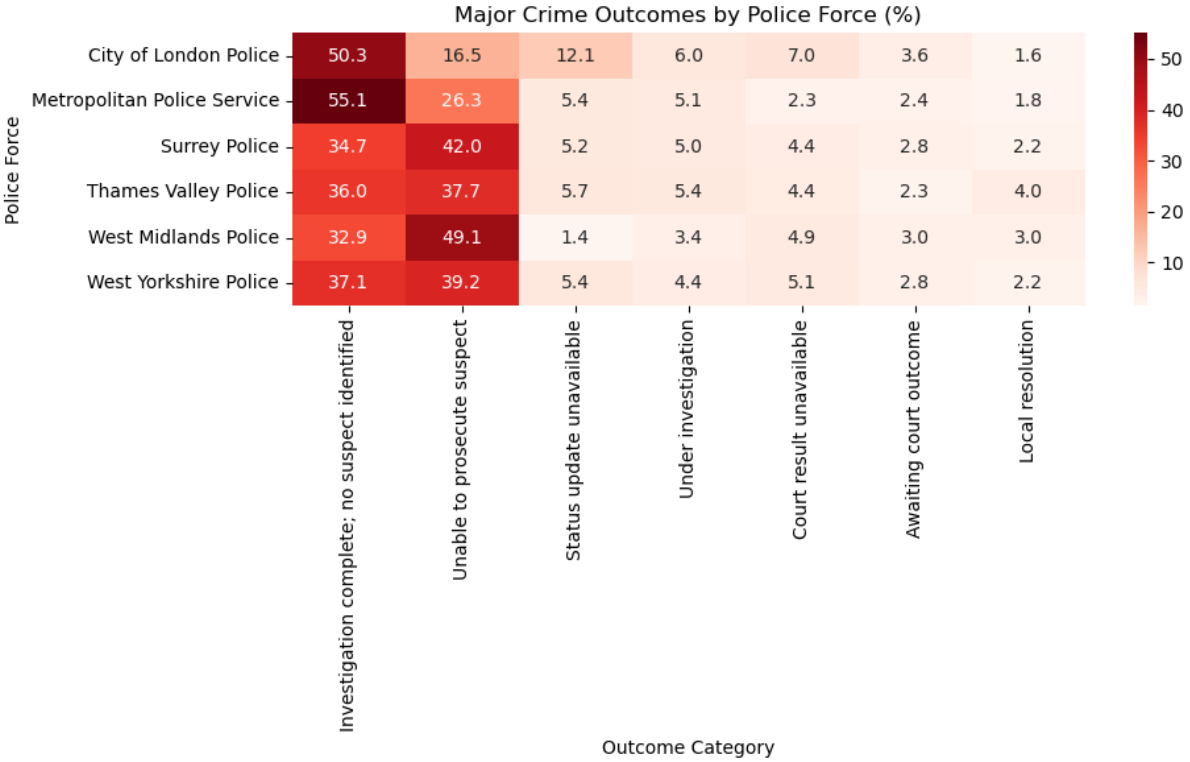
The second most common outcome is "Unable to prosecute suspect/" This outcome indicates that although a suspect may have been identified, there was insufficient evidence to pursue a legal prosecution.

Several other outcomes appear at much lower frequencies, including "Under investigation", "Court result unavailable", and "Awaiting court outcome". These represent cases that are still progressing through the criminal justice system.

Outcomes such as "offender given a caution", "penalty notice", or 'local resolution" represent a relatively small share of cases, indicating that only a limited proportion of recorded crimes result in formal enforcement actions.

Overall, the distribution highlights that a large proportion of reported crimes do not result in a confirmed suspect of formal legal outcome, which provides important context when interpreting crime statistics.

7.2. Crime Outcome percentage in each Police Force Jurisdictions



The data presents a heatmap showing the percentage distribution of major crime outcomes across police forces, allowing comparison of how cases are resolved in different regions. Across most police forces, “Investigation complete; no suspect identified” remains the most common outcome, particularly for the Metropolitan Service and the City of London Police. This suggests that identifying suspects may be more difficult in highly urban environments where crime volumes are larger, and incidents may involve unknown offenders.

However, there are notable regional differences in outcome distribution. For example, West Midlands Police show a particularly high proportion of cases classified as “Unable to prosecute suspect,” indicating that although suspects may be identified, prosecution is often not pursued due to insufficient evidence.

Police forces such as Surrey Police and Thames Valley Police show slightly more balanced outcome distributions, with somewhat lower proportions of cases ending without a suspect identified and a slightly higher share of cases progressing further in the justice process.

The heatmap also highlights that local resolution outcomes remain relatively small across all regions, suggesting that informal resolutions represent only a minor share of crime outcomes in the analysed dataset.

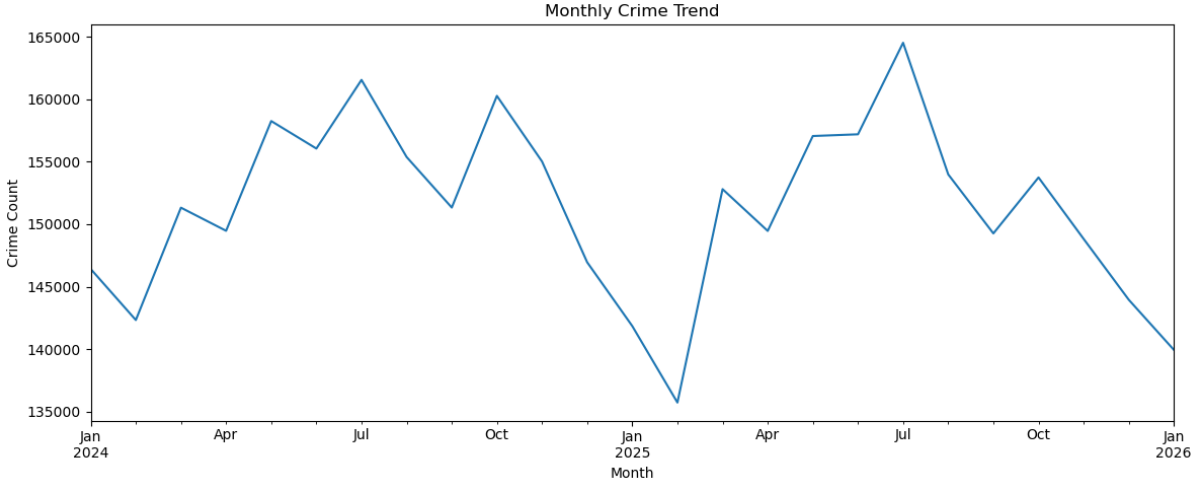
7.3. Crime Outcome - Key Insights

These two figures show that while crime volumes differ between police forces, the majority of cases across all regions result in outcomes where no suspect is identified, or prosecution cannot proceed. This suggests that the effectiveness of case resolution may be influenced by factors such as evidence availability, reporting circumstances, and the complexity of certain crime types.

Understanding crime outcomes provides additional context to crime statistics, as regions with similar crime levels may still differ in how cases progress through the criminal justice process. These findings also provide additional context for regional crime comparisons, as crime outcomes may influence public perceptions of safety and confidence in law enforcement.

8. Exploratory Data Analysis – Time Analysis

8.1. Monthly Crime Trend Across Two Years



This graph shows the total number of recorded crimes, aggregated monthly across all analysed police forces.

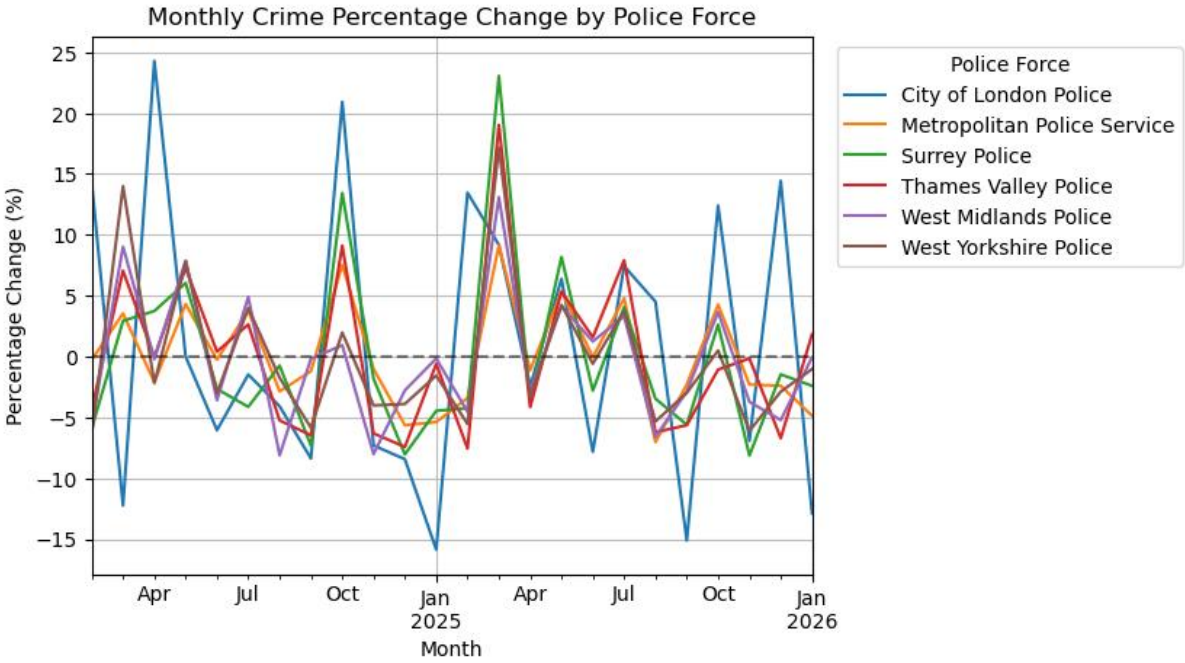
The chart shows that crime levels fluctuate moderately over time but generally remain within a relatively stable range. Monthly crime counts typically range from approximately 140,000 to 165,000 incidents, indicating that while crime levels fluctuate from month to month, there are no extreme long-term increases or decreases over the analysed period.

A noticeable dip in crime levels occurs during early 2025, when recorded incidents drop to the lowest point in the dataset. This may reflect seasonal variation, reporting differences, or short-term fluctuations in crime activity.

Following this decline, crime levels increase again during mid-2205, reaching a peak around June-July 2025, which represents the highest point in the time series. This suggests that crime activity may increase during summer months, potentially due to higher levels of social activity, tourism, and outdoor movement.

Overall, the time series suggests moderate seasonal variation, with crime levels tending to increase during warmer months and decline slightly during winter periods.

8.2. Monthly Crime Percentage Change in Each Police Jurisdiction



The figure above shows the month-to-month percentage change in crime levels for each police force. This allows comparison of how crime trends fluctuate across different regions. Across all police forces, the percentage changes generally fluctuate within a relatively moderate range of approximately ± 10%, suggesting that crime levels tend to change gradually rather than experiencing extreme monthly volatility.

Several police forces experience occasional larger increases or decreases, mainly around the beginning of 2025, where some regions show sharper declines followed by recovery in subsequent months. These variations may reflect temporary changes in reporting patterns, seasonal behaviour, or specific regional events.

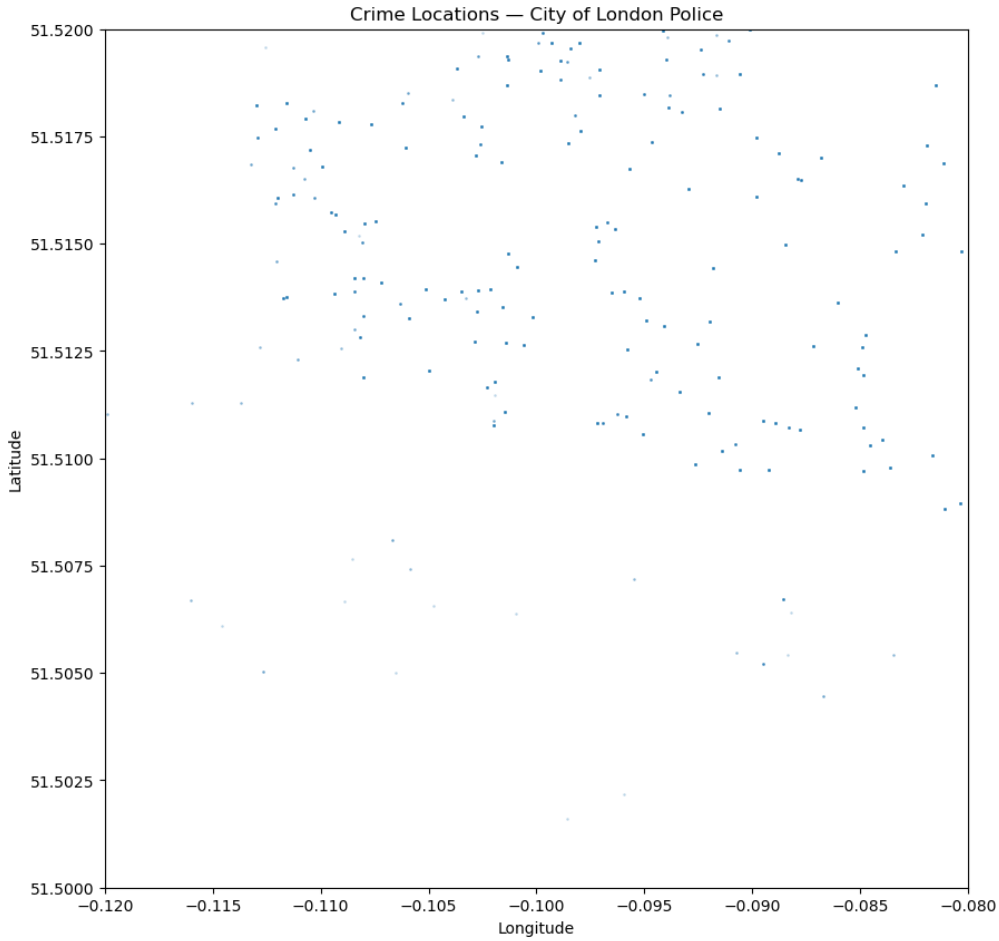
Despite these fluctuations, the overall pattern across the police force remains relatively consistent, with most regions showing similar directional movements in crime levels over time. This suggests that broader social or seasonal factors may influence crime trends across multiple regions simultaneously.

8.3. Time Analysis - Key Insights

These figures indicate that crime levels show moderate seasonal variation but remain relatively stable overall during the analysed period. While month-to-month fluctuations occur across all police forces, there is no evidence of sustained long-term growth or decline in crime levels. Analysing the time trend provides useful context for interpreting crime statistics, as certain months may experience higher or lower crime activity due to seasonal factors or a change in public behaviour.

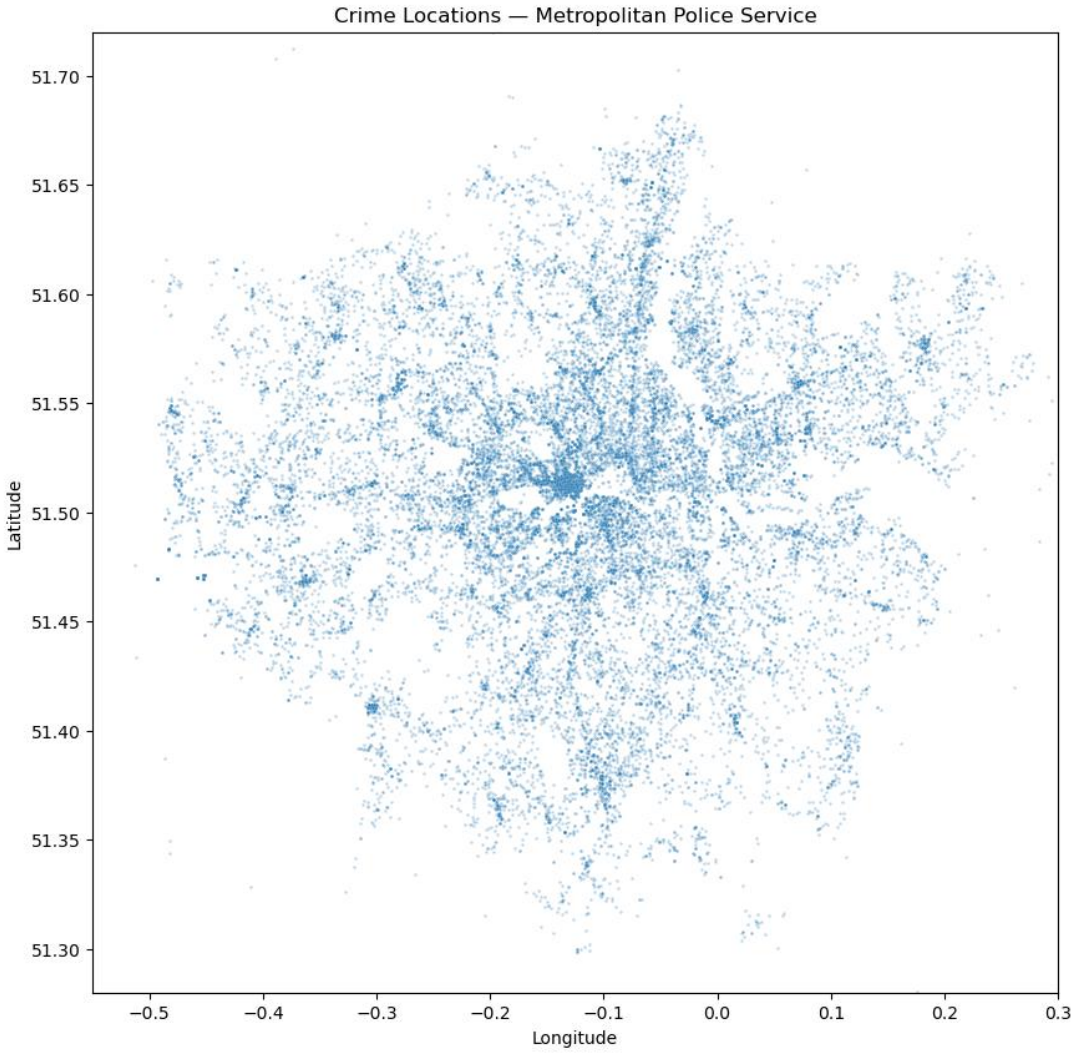
9. Exploratory Data Analysis – Crime Map

9.1. City of London



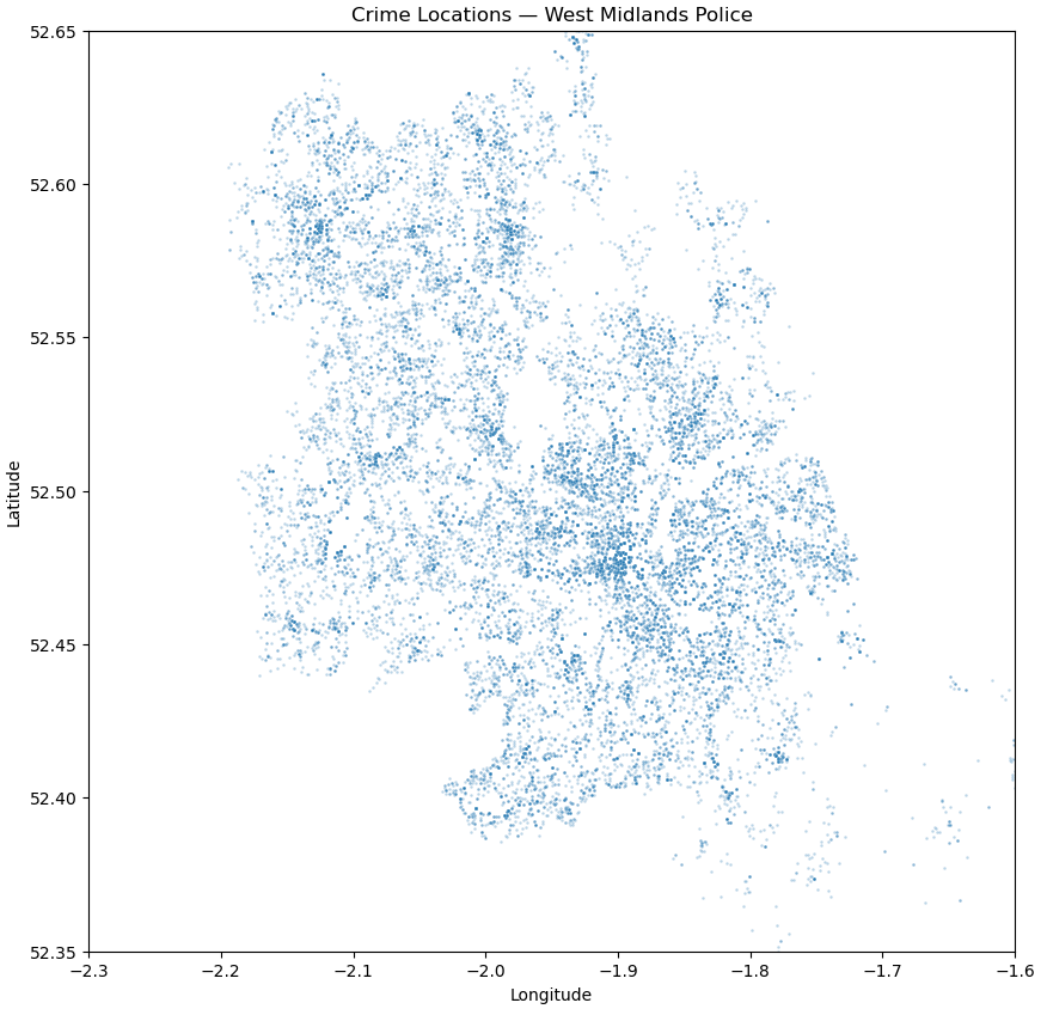
The City of London map shows that there are very few concentrated crime incidents clustered, which supports the crime volume of the region. Crime locations appear concentrated around the central north of the River Thames, where high levels of commercial activity and tourism may contribute to increasing opportunities for certain crime types such as theft and shoplifting.

9.2. Metropolitan Police Service



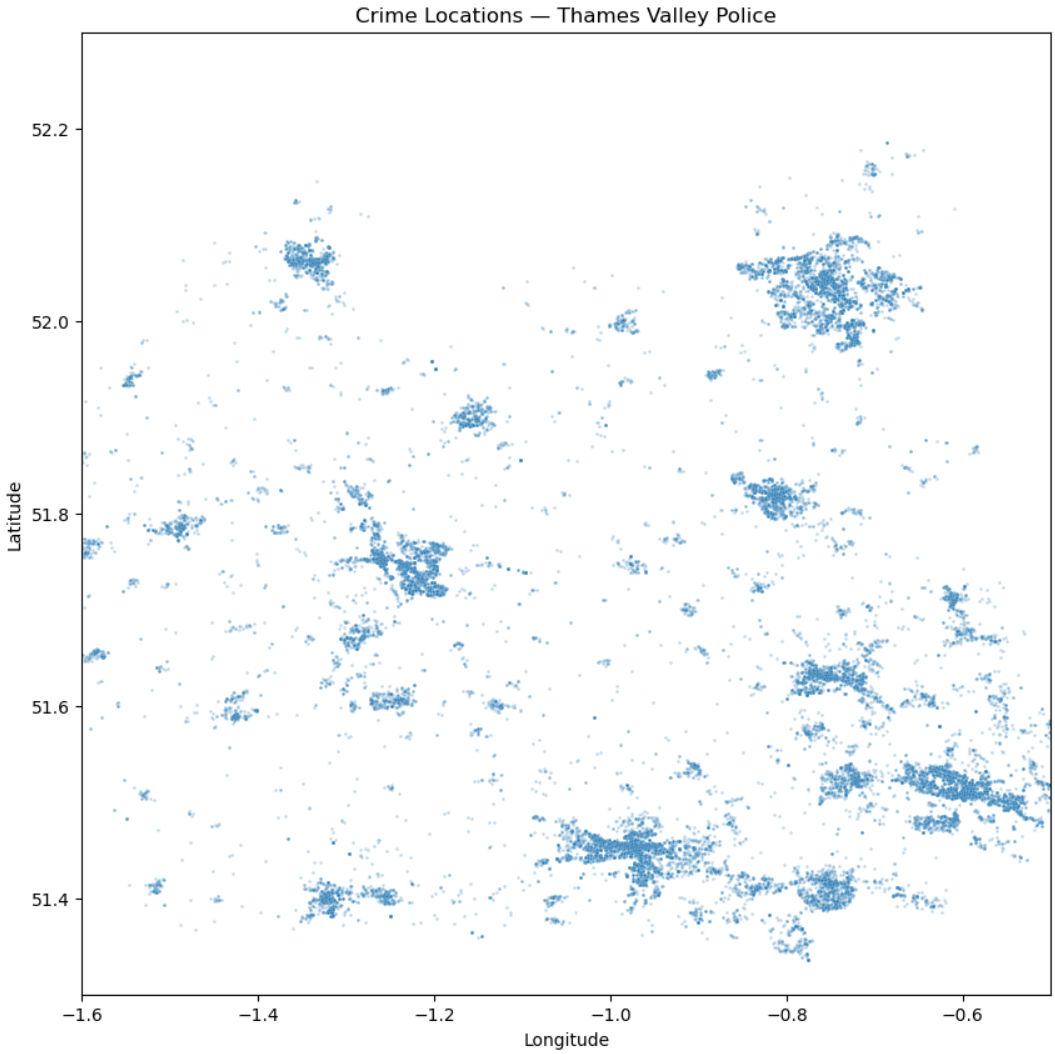
The Metropolitan Police map shows a dense and extensive distribution of crime incidents across Greater London. Crimes are heavily concentrated in central areas, with clusters extending outward along major urban corridors. The high density of points reflects the large population and high level of economic and social activity within the capital.

9.3. West Midlands Police



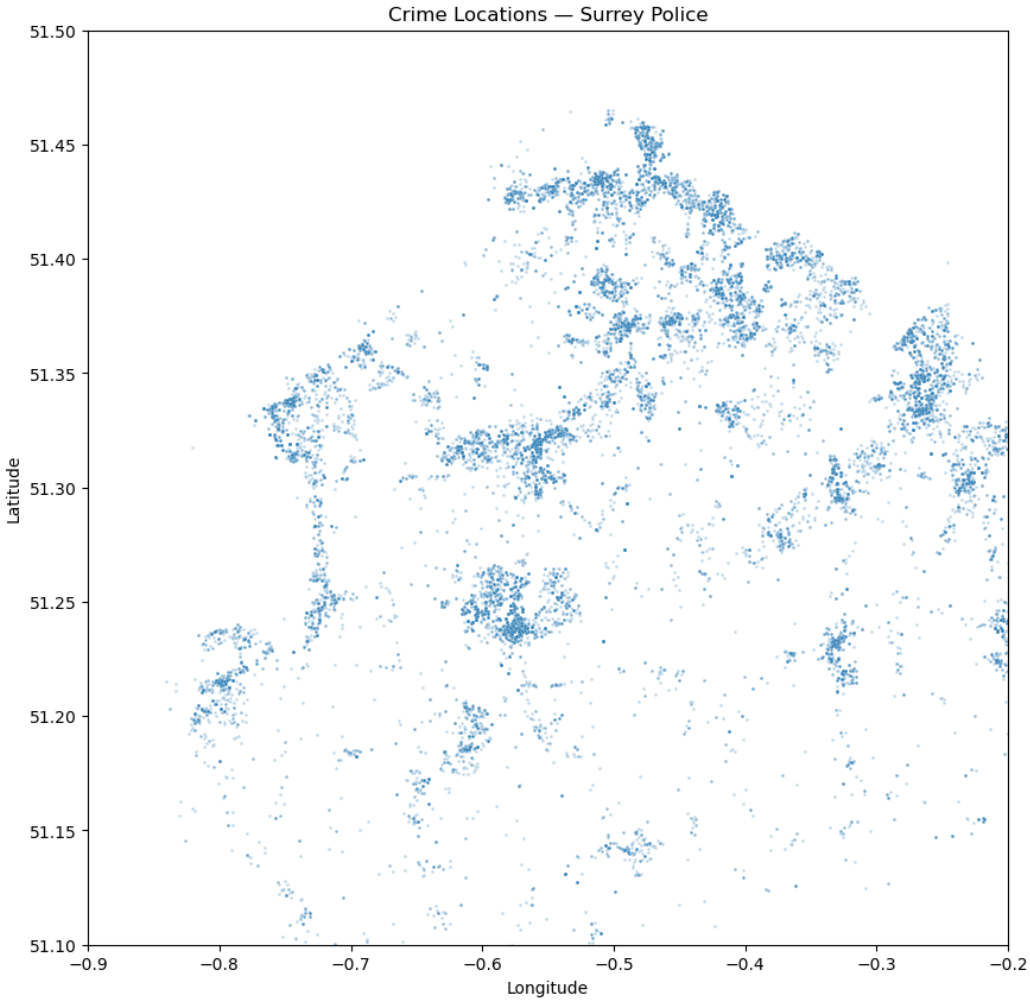
The spatial distribution of crime in the West Midlands region shows several clusters concentrated around major urban centres, particularly the Birmingham metropolitan area. Crime incidents are distributed across multiple neighbouring districts, indicating that crime activity is spread across several densely populated urban areas rather than a single focal point.

9.4. Thames Valley Police



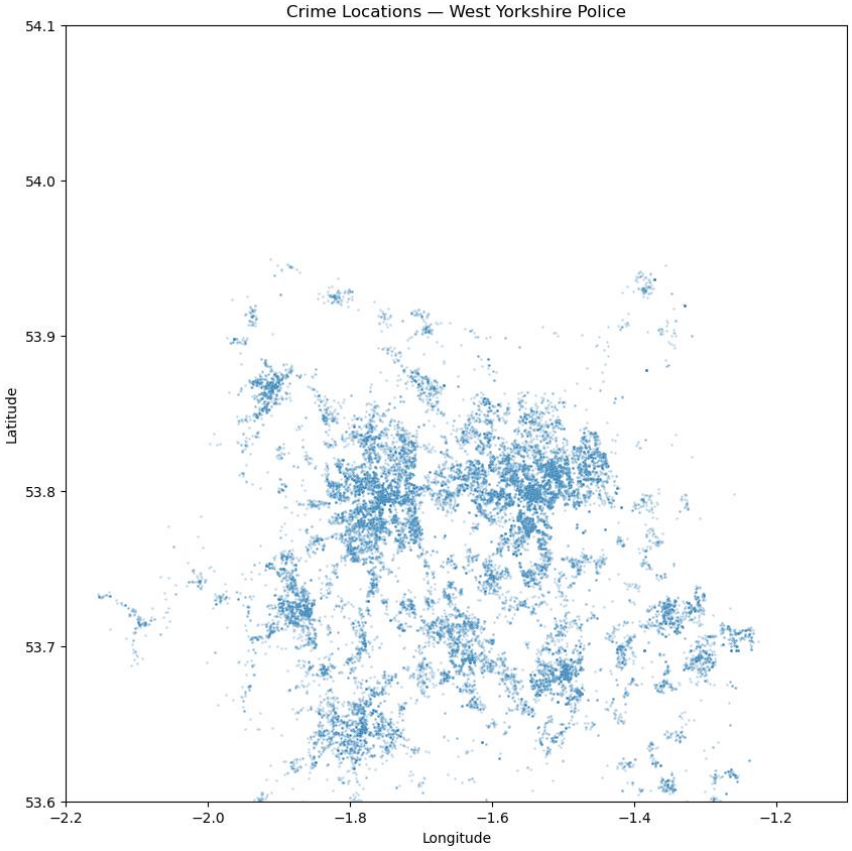
The Thames Valley Police map shows multiple distinct clusters of crime across the region. These clusters likely correspond to major towns and urban centres within the Thames Valley area. Compared with highly urban regions such as London, crime incidents appear more dispersed, reflecting a mixture of suburban and semi-rural environments within this police force area.

9.5. Surrey Valley Police



Crime locations within the Surrey Police area appear more spatially dispersed, just like other suburban areas, with several smaller clusters spread throughout the region. This pattern is consistent with Surrey's predominantly suburban and semi-rural landscape, where crime incidents tend to occur around town centres rather than forming a single large urban concentration.

9.6. West Yorkshire Police



The West Yorkshire Police map shows several distinct clusters of crime concentrated around major urban centres, particularly in areas corresponding to Leeds, Bradford, and surrounding towns. Crime incidents appear densely concentrated in these urban areas, while more rural parts of the region show lower incident density. This spatial distribution reflects the urban structure of West Yorkshire, where several large towns contribute to regional crime patterns rather than a single dominant city centre.

9.7. Key Spatial Insight

Across all analysed police forces, crime incidents show clear geographic clustering around urban centres and densely populated areas. Regions with large metropolitan populations, such as the Metropolitan Police, West Midlands Police, and West Yorkshire Police, exhibit the highest concentration of crime locations, reflecting the influence of population density, economic activity, and urban infrastructure.

In contrast, regions with more suburban or semi-rural characteristics, such as Surrey and parts of the Thames Valley, show more dispersed crime patterns with smaller clusters around local town centres.

These spatial differences highlight how urbanisations and population distribution strongly influence crime patterns. Understanding where crimes are geographically concentrated provides valuable context for comparing regions and supports the broader objective of identifying police forces that may warrant further investigation in the next stage of the project.

10. Key Insights

The exploratory data analysis examined crime patterns across six UK police forces using multiple analytical perspectives, including total crime counts, crime rates relative to population, crime type distributions, crime outcomes, temporal trends, and spatial crime locations.

The analysis showed substantial differences in overall crime volumes across regions. The Metropolitan Police Service recorded the highest number of total crimes, which is expected, given the large population and high population density of Greater London. In contrast, smaller jurisdictions such as the City of London Police recorded far fewer incidents due to their limited geographic coverage.

However, when adjusting crime counts relative to population size, the ranking of police forces changes significantly. Regions such as West Yorkshire Police and West Midlands Police show higher crime rates per 1000 residents compared with other areas. This demonstrates the importance of normalising crime data when comparing regions with very different population sizes.

Analysis of crime types shows that violence and sexual offences represent the largest share of crimes across all police forces. Theft-related crimes, including shoplifting and other theft, also contribute significantly to the overall crime landscape. Despite variations in total crime levels, the general composition of crime types remains relatively consistent across regions.

The analysis of crime outcomes indicates that a large proportion of recorded crimes result in outcomes where investigations are completed without identifying a suspect or where suspects cannot be prosecuted. This suggests that many reported incidents do not lead to formal legal action, which provides important context when interpreting crime statistics.

Examining crime trends over time reveals moderate monthly fluctuations but no clear long-term increase or decrease in crime levels during the analysed period. Crime levels tend to vary within a relatively stable range, with occasional peaks during mid-year months and lower levels during winter periods.

Finally, the spatial analysis of crime locations highlights clear geographic clustering patterns. Crime incidents tend to concentrate in urban centres with higher population density and economic activity. Metropolitan areas such as London, Birmingham, and Leeds show particularly dense clusters of crime locations, while more suburban regions exhibit more dispersed spatial patterns.

Overall, the exploratory analysis demonstrates that crime patterns vary across regions in terms of crime volume, crime rate, and geographic distribution, while maintaining broadly similar crime type compositions. These findings provide a comprehensive overview of crime across the analysed police forces and establish a strong foundation for identifying police force areas that warrant further investigation in the next stage of the project.

11. Recommendations

Based on the exploratory data analysis, it is recommended that the Metropolitan Police Service and West Midlands Police be selected for further investigation.

The Metropolitan Police Service recorded the highest overall number of crimes across the dataset and showed the most extensive spatial clustering of crime incidents. The geographic visualisation indicates a very dense concentration of crime across Greater London, particularly in central urban areas. Given London's large population, high economic activity, and significant crime volume, further investigation of this police force could provide valuable insights into the drivers of crime in major metropolitan environments.

The West Midlands Police area is also recommended for further analysis due to its relatively high crime rate and clear clustering of crime locations around major urban centres. The spatial analysis suggests multiple crime hotspots across the region, likely corresponding to cities such as Birmingham and surrounding districts. Compared with London, the West Midlands represents a different type of urban environment with several clustered population centres, making it an interesting region for comparative analysis.

By focusing on these two police forces, the analysis can explore how crime patterns differ between the UK's largest metropolitan policing area and another major urban region. Investigating these areas further may help identify patterns in crime concentration, crime types, and temporal trends that could support more targeted policing strategies.

12. Limitations

First, the analysis is based on reported crime data, which may not represent the full extent of criminal activity. Some crimes may go unreported, meaning that the dataset may underestimate the true level of crime in certain regions.

Second, the analysis primarily uses crime counts and crime rates relative to population, which may not fully capture differences in daytime population, tourism, or commuting patterns. For example, areas such as the City of London have relatively small residential populations but large daily working populations, which may affect crime rate calculations.

Third, the dataset provides limited contextual information about individual crimes. Factors such as socio-economic conditions, policing strategies, and environmental characteristics are not included in the dataset but may influence crime patterns.

Fourth, the spatial analysis is based on latitude and longitude coordinates of crime locations, which represent approximate locations rather than exact incident sites. This may slightly affect the precision of geographic crime clustering visualisations.

Finally, the analysis covers a limited time period of two years, which may not fully capture longer-term crime trends or structural changes in crime patterns.

Despite these limitations, the exploratory analysis still provides valuable insights into regional crime patterns and supports the identification of police forces for further investigation.